

Advancing Brain-Computer Interface (BCI) Technology

As a Master's student in Microelectronics Technology and Materials at The Hong Kong Polytechnic University, I have cultivated a multidisciplinary skill set spanning image processing, AI-based image recognition, and electronics engineering. My decade-long professional experience includes roles such as Algorithm Engineer and R&D Lead in notable organizations like OPT and SmartMore Technology. These experiences equipped me with expertise in software development (C++ and Python), model optimization, data preprocessing, and system integration. Additionally, I hold over ten invention patents and have a strong foundation in designing and prototyping Bluetooth communication devices and robotic systems.

Brain-Computer Interface (BCI) technology has been a long-standing passion of mine, representing a confluence of engineering, materials science, and neuroscience. This interest has guided my academic and professional pursuits, inspiring my transition from software and AI applications to semiconductor materials and microphysics. By addressing the challenges of electrode precision and signal acquisition, I aim to advance the field of BCI, which I believe holds transformative potential in healthcare, rehabilitation, and human-computer interaction.

To align my expertise with the demands of BCI research, I have focused on strengthening my interdisciplinary knowledge. My Master's research has enabled me to delve into nanomaterials and microphysics, providing essential insights into the interaction between biological signals and electrode materials. This foundation has prepared me to investigate advanced signal acquisition techniques, refine AI-driven signal recognition models, and explore the integration of BCI with robotics—particularly in developing brain-controlled prosthetics.

If admitted to CUHK's PhD program, I aim to contribute to its vibrant research ecosystem through the following objectives:

1. **Data Acquisition and Signal Analysis:** Collect and preprocess large-scale neural datasets, ensuring data quality through advanced cleaning and classification methods.
2. **AI Model Development:** Design robust neural signal recognition models to improve accuracy and stability, leveraging CUHK's strengths in AI research.
3. **Real-World Applications:** Investigate practical implementations of BCI, particularly in robotics, to create innovative rehabilitation tools.
4. **Interdisciplinary Collaboration:** Work closely with neuroscientists, material scientists, and medical professionals to advance electrode technologies and enhance BCI precision.

I am acutely aware of the challenges in this field, including the low initial accuracy of signal recognition models and the variability in device performance across environments. My professional background has taught me the importance of addressing such challenges through data quality enhancement, precision engineering,

and cross-disciplinary teamwork. By gaining deeper insights into neuroscience and medical science, I am committed to bridging the gap between technology and healthcare solutions.

CUHK's pioneering research in neuroscience and biomedical engineering aligns perfectly with my aspirations. Its collaborative and innovative research environment provides an ideal platform to advance BCI technology. I look forward to contributing to CUHK's research community and making a meaningful impact in this transformative field.